Problem F
Question 2: Generalized Scaling Law
Explicit Q1 injection, quality scenarios, attenuation, elasticities and substitution

Final synchronized technical report

24 September 2026

Numerical source of truth. Every number, table and plotted quantity in this report is synchronized to the latest successful `python run_all.py` execution supplied by the team. Older report values are superseded.

Abstract

The latest Q2 pipeline explicitly executes the Q1-to-B-series bridge rather than treating the quality coordinate as independent. B-series quality ranks are mapped through the A1 empirical CDF to an A1 Q_i value and then compressed by a sigmoid. This changes the fitted quality law materially: the main B6/B7 scenario uses $G = 1.726229$, $\gamma = 3.351048$, while the reverse B8 scenario uses $H = 6.664889$, $\delta = 2.725690$. The B1 scaling law and Hill attenuation remain unchanged. The new report also includes the executable ECDF bridge table, loss elasticities, and conditional quality-to-model-size substitution calculations.

Contents

1 Classical scaling law **1**

2 Explicit Q1-to-Q2 injection **1**

2.1 Three-level direction diagnostic 2

3 Structural and transfer validation **3**

4 Scale-dependent mixture attenuation **3**

5 Final generalized macro law **3**

6 Elasticities and substitution **4**

7 Known limitations - honest disclosure **5**

8 Final Q2 answer **5**

1 Classical scaling law

With N, D in billions,

$$L_0(N, D) = E + AN^{-\alpha} + BD^{-\beta},$$

where

$$(E, A, \alpha, B, \beta) = (1.689798, 0.353980, 0.339977, 1.240306, 0.279878).$$

The B1 fit has $R^2 = 0.99999982$, MAE $= 1.052 \times 10^{-4}$ and RMSE $= 1.465 \times 10^{-4}$. A trajectory-cluster bootstrap over eight N trajectories yields 2,000 successful draws.

Table 1: B1 point estimates and 95% trajectory-cluster intervals.

Parameter	Estimate	2.5%	97.5%
E	1.689798	1.689666	1.689952
A	0.353980	0.353842	0.354115
α	0.339977	0.339837	0.340115
B	1.240306	1.240161	1.240433
β	0.279878	0.279827	0.279937

2 Explicit Q1-to-Q2 injection

The latest executable bridge is

$$Q_B \longrightarrow u \longrightarrow Q_i^{A1} = F_{A1}^{-1}(u) \longrightarrow Q_\sigma = \sigma(Q_i^{A1}) \in (0,1).$$

The persisted bridge has 21 percentile knots from 0 to 1. For example, $u = 0.10 \mapsto Q_i = -0.504235$, $u = 0.50 \mapsto 0.026702$, and $u = 0.90 \mapsto 0.458469$.

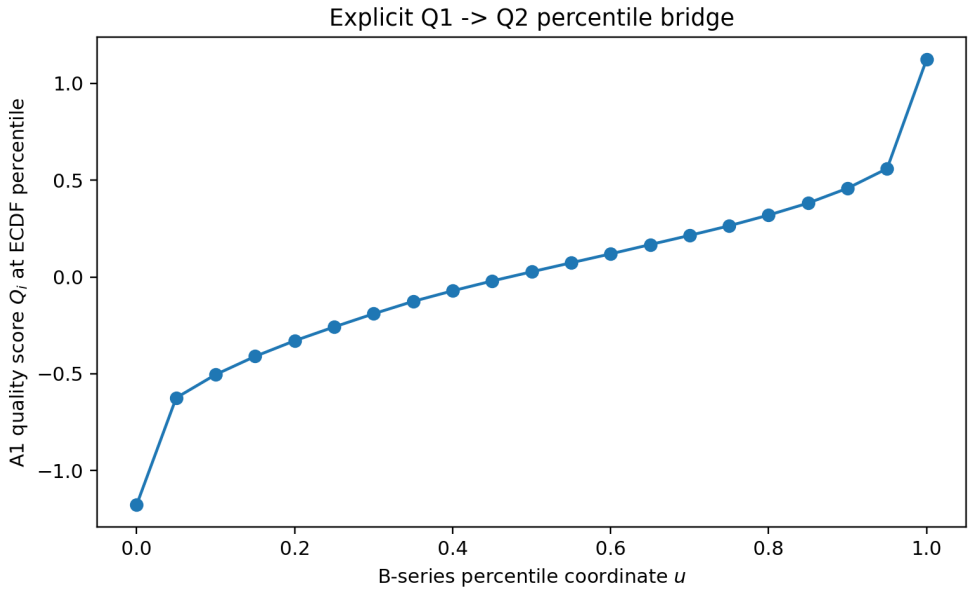


Figure 1: A1 ECDF quantile map used in the explicit Q1 injection.

For B6, the injected Q_i^{A1} range is $[-0.504, 1.124]$, which becomes $Q_\sigma \in [0.377, 0.755]$ with median 0.506. The main quality law is

$$\Delta_Q^{(+)}(Q) = 1.726229(1-Q)^{3.351048}.$$

The fit has $R^2 = 0.9465$, MAE $= 0.06245$, and 95% intervals

$$G \in [1.500910, 1.960607], \qquad \gamma \in [3.179723, 3.550433].$$

On the 90 B7 rows absent from B6, the frozen fit gives $R^2 = 0.9638$, MAE $= 0.04648$, Spearman $= 0.9893$.

B8 is modeled separately:

$$\Delta_Q^{(-)}(Q) = -6.664889(1-Q)^{2.725690},$$

with $R^2 = 0.8661$, MAE $= 0.21121$, $H \in [6.451453, 6.879929]$, and $\delta \in [2.680189, 2.770978]$.

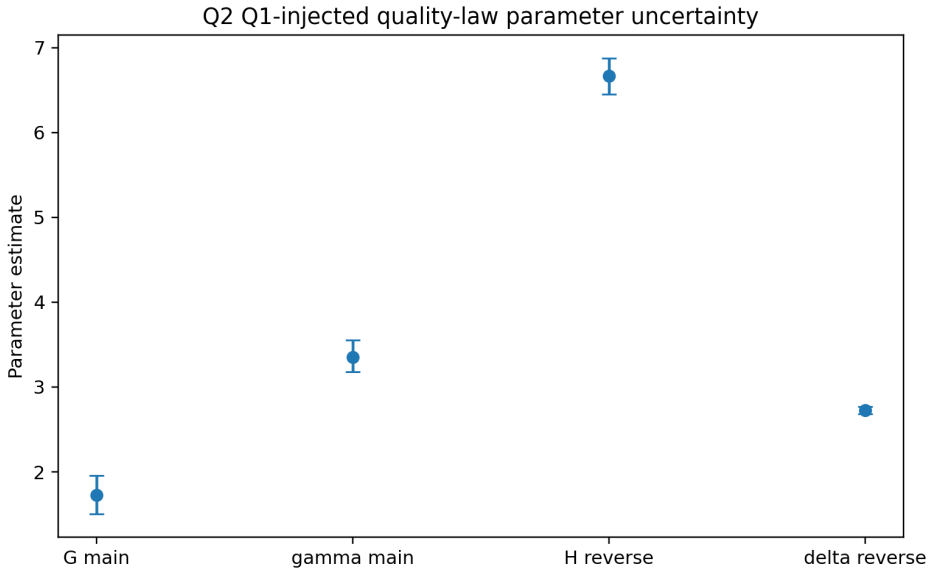


Figure 2: Uncertainty in the Q1-injected quality-law parameters.

2.1 Three-level direction diagnostic

Within-series Spearman is the primary sign evidence: B6 has $\rho(Q_\sigma, L) = -0.3239$ and B8 has $+0.9208$, matching their intended directions. The sigmoid widths are only 0.378 (B6) and 0.406 (B8). Cross-series matched correlation is $\rho = -0.2590$ on 160 matched points; because the quality range is compressed and B6/B8 are independent generating runs, this L3 statistic is treated as secondary rather than as the primary direction test.

3 Structural and transfer validation

B2 independently fit with the same form gives $(E, A, \alpha, B, \beta) = (1.669638, 0.360298, 0.335765, 8.525028, 0.277357)$ and $R^2 = 0.999545$. Direct B1 transfer to B4 gives $R^2 = 0.6045$, MAE 0.2273, Spearman 0.9830; B5 gives $R^2 = 0.7306$, MAE 0.1707, Spearman 0.9588. B10 gives $R^2 = 0.999992$, MAE 0.000614, Spearman 0.999994, but it is an **estimated** 100B+ consistency check rather than independent empirical validation.

4 Scale-dependent mixture attenuation

Relative mixture-sensitivity factors are 1.000 (1M), 0.871 (60M), 0.425 (1B), 0.260 (10B estimated), and 0.192 (70B estimated). They are smoothed by

$$\omega(N) = \omega_\infty + (1 - \omega_\infty)\frac{1 + (N_0/N_c)^\eta}{1 + (N/N_c)^\eta}, \quad N_0 = 0.001,$$

with

$$\boxed{\omega_\infty = 0.198813, \quad N_c = 0.360619 \text{ B}, \quad \eta = 0.879637.}$$

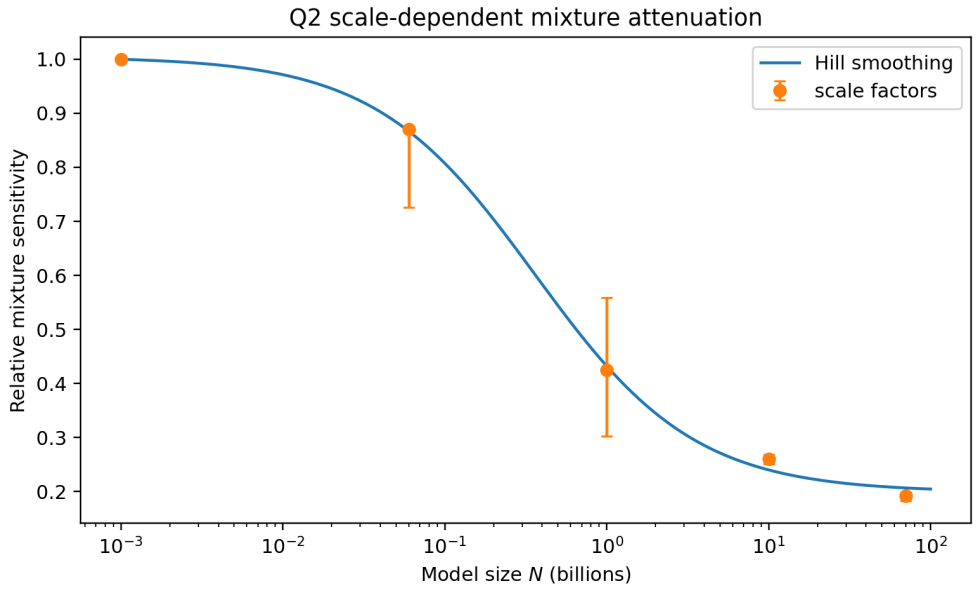


Figure 3: Latest Hill attenuation; 10B and 70B anchors are estimated.

5 Final generalized macro law

The main scenario is

$$\boxed{L^{(+)} = 1.689798 + 0.353980N^{-0.339977} + 1.240306D^{-0.279878} + 1.726229(1-Q)^{3.351048} + \omega(N)\Delta_p(p).}$$

The reverse scenario replaces the quality term by $-6.664889(1-Q)^{2.725690}$; the conservative scenario uses no quality term. This is a macro-average law.

6 Elasticities and substitution

Table 2: Latest loss elasticities from the executable bridge-and-elasticities stage.

Point	N	D	Q	ϵ_N	ϵ_D	ϵ_Q
Small	0.1	10	0.5	0.080157	0.055482	0.172603
Medium	1	100	0.5	0.047106	0.037445	0.221904
Large	10	1000	0.7	0.026684	0.024359	0.049646
Very large	100	3000	0.8	0.013210	0.019398	0.013818

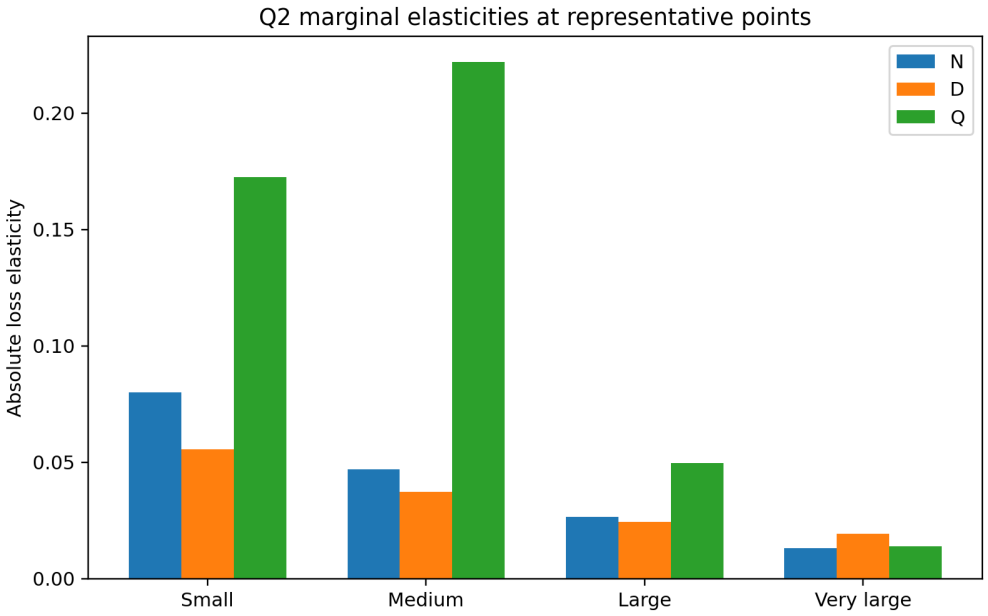


Figure 4: Absolute elasticities at representative points.

A +0.1 change in the injected quality coordinate has the following conditional parameter equivalences:

Table 3: Conditional quality-to-model-size substitution.

N_0	D	Q_0	ΔQ	N_{eq}	N_{eq}/N_0
0.1	50	0.5	0.1	0.143256	1.4326
1	200	0.6	0.1	1.558212	1.5582
5	500	0.7	0.1	7.062990	1.4126
20	1000	0.8	0.1	23.648840	1.1824

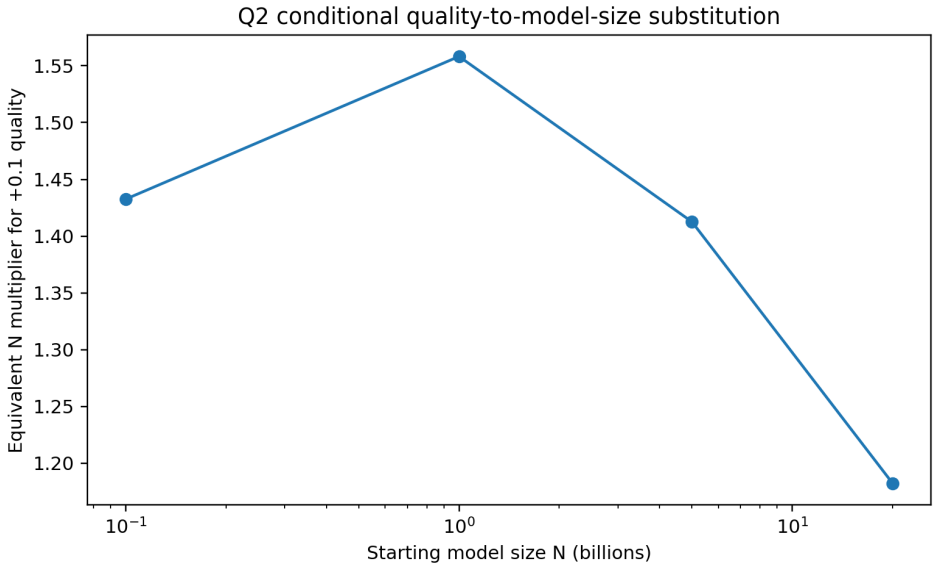


Figure 5: Conditional equivalent model-size multiplier for +0.1 quality.

7 Known limitations - honest disclosure

Q2 limitations. (i) The Q1-to-B bridge is executed, but the sigmoid compresses the injected quality range to widths about 0.38–0.41, weakening cross-series ordering. Primary direction evidence is within-series L1, not the cross-series L3 correlation. (ii) The percentile/ECDF bridge imposes a chosen cardinalization of an originally proxy-like score; elasticities and substitution rates are conditional on that mapping. (iii) The 10B and 70B attenuation anchors are estimated. (iv) B10 is an estimated 100B+ extrapolation consistency check, not measured validation. (v) B8 contradicts the B6/B7 direction, so Q3 must carry multiple quality-evidence scenarios.

8 Final Q2 answer

The scaling exponents remain stable, but the quality term changes materially once the Q1 injection is actually executed. This is the version that must feed Q3: $G = 1.726229$, $\gamma = 3.351048$, $H = 6.664889$, $\delta = 2.725690$, together with the unchanged Hill attenuation and Δ_p response.